

AAPG2025	BENEFIT		PRC
Coordinated by	Lionel Larchevêque	48 months	634,728.32 €
CES 46, Axe E.5 – “Calcul haute performance, Modèles numériques, simulation, applications”			

BENEFIT

Stability-integrated machine Learning For active flow control

Consortium: IUSTI, Pprime, DynFluid

Summary table of persons involved in the project

Partner	Last name	First name	Current position	Role & responsibilities	Involvement (pers.month)
IUSTI	LARCHEVÊQUE	Lionel	MCF	Coordinator Partner coordinator Leads WP1, WP0 Co-supervision PhD1 Inv. Tasks 1.1, 1.3, 3.2b	18 p.m
IUSTI	DUPONT	Pierre	CR CNRS - HDR	Co-supervision PhD2 Inv. Tasks 1.1, 1.3, 3.2b	8 p.m
IUSTI	PIPONNIAU	Sébastien	MCF	Inv. Tasks 1.1, 1.3, 3.2b	8 p.m
IUSTI	LANTOINE	Pierre	IE	Wind tunnel operations Inv. Tasks 1.1, 3.2b	8 p.m
IUSTI	To be recruited	-	PhD Student	Tasks 1.1, 1.2, 3.1, 3.2b Co-supervised by P. Dupont and P. Traoré	36 p.m
Pprime	AGOSTINI	Lionel	CR CNRS - HDR	Partner coordinator Leads WP3 Co-supervision PhD3 Inv. Tasks 1.2, 2.2, 2.3, 3.1, 3.2a	10 p.m
Pprime	TRAORÉ	Philippe	PR	Co-supervision PhD2 Inv. Tasks 1.1, 3.1, 3.2b	10 p.m
Pprime	CORDIER	Laurent	DR CNRS	Inv. Tasks 2.2, 2.3, 3.1, 3.2a, 3.2b	8 p.m
Pprime	KERHERVÉ	Franck	MCF	Inv. Tasks 1.2, 3.2b	8 p.m
Pprime	To be recruited	-	PhD Student + Master	Tasks 2.2, 2.3, 3.2a Co-supervised by J.-C. Loiseau and L. Agostini	36+6 p.m
DynFluid	ROBINET	Jean-Christophe	PR	Partner coordinator Leads WP2 Co-supervision PhD1 Tasks 1.1, 1.3, 2.1	10 p.m
DynFluid	LOISEAU	Jean-Christophe	MCF	Co-supervision PhD3 Tasks 2.2, 2.3, 3.2a	8 p.m
DynFluid	ALFEREZ	Nicolas	MCF	Numerical code development Tasks 1.1, 1.3, 2.1	8 p.m
DynFluid	To be recruited	-	PhD Student + Master	Tasks 1.1, 1.3, 2.1, 3.2a Co-supervised by J.-C. Robinet and L. Larchevêque	36+6 p.m

Changes made in the full proposal compared to the pre-proposal

The full proposal includes a significant enhancement to our research framework that was not present in the pre-proposal. Originally, we requested funding for a postdoctoral researcher to address specific aspects of WP2 and WP3. After careful consideration and further development of our methodology, we have introduced Task 3.2a, which represents a more ambitious and generalised approach to achieving the project's objectives.

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Given the longer-term nature of this challenging research direction and its alignment with doctoral training opportunities, we have decided to request PhD funding rather than postdoctoral support. This adjustment allows for a more sustained investigation of the ambitious aims in Task 3.2a while simultaneously providing valuable training in this emerging interdisciplinary field.

Task 3.2a focuses on developing reduced-order models with embedded control effects, enabling dynamic adaptation of control strategies to varying objectives without requiring additional simulation or experimental data. This approach complements the more conservative Task 3.2b (data assimilation and reinforcement learning for real-time adaptation), which was already included in the pre-proposal.

This dual-track strategy offers two majors advantages. Firstly, it mitigates research risk by pursuing parallel approaches with different risk profiles—Task 3.2a represents a more challenging but potentially transformative method, while Task 3.2b provides an easiest pathway to ensure project deliverables. Secondly, the addition of Task 3.2a significantly enhances the scope and impact of the research by aiming for control laws that can be adapted to new objectives "on the fly" without retraining.

We believe that this enhancement substantially strengthens the proposal by balancing ambitious research goals with practical implementation considerations, thereby increasing both the scientific impact and the likelihood of successful outcomes.

I Proposal’s context, positioning and objective(s)

This research proposes a novel framework that synergistically integrates **model-driven** stability analysis with **data-driven** machine learning techniques to design interpretable control strategies for fluid flows. Current approaches to flow control face significant limitations: stability-based methods provide valuable physical insights but are inherently limited in capturing the nonlinear dynamics essential for practical control applications, whilst data-driven approaches often confront difficulties with the high-dimensional and multi-scale characteristics inherent in turbulent flows.

a Objectives and scientific hypotheses

This research aims to develop a comprehensive framework for designing active **flow control strategies** that harness **inherent flow instabilities** through the **coupling of model-driven and data-driven methods**. The central hypothesis is that by exploiting the natural instabilities and strategically influencing their dynamics, it is possible to optimise specific quantities such as friction, heat transfer or other relevant parameters with minimal energy input. This approach offers particular advantages where maintaining control effectiveness at high flow velocities can become energy-intensive. Although the framework accommodates various active actuators, plasma actuators form the focus of the current project. Despite their versatility in enabling sophisticated spatio-temporal control strategies, plasma actuators have seen limited adoption in real-world applications due to the relatively modest magnitude of perturbations they generate. The proposed framework addresses this limitation by specifically targeting flow instabilities, where even small perturbations can trigger significant flow modifications.

To achieve these objectives, **several scientific and technical barriers** must be overcome. **Model-driven approaches** based on **stability analysis** face limitations due to their reliance on linearised equations, which can deviate from actual nonlinear flow behaviour. Traditional **machine learning methods** encounter difficulties with the high-dimensionality of turbulent flows, often producing non-interpretable **"black-box"** models. Additionally, numerical simulations require substantial computational resources, limiting the range of configurations that can be investigated.

The innovative aspect of this proposal lies in the integration of stability analysis with machine learning algorithms that exploit empirical flow data. **Stability analysis** serves as an effective method for identifying flow instabilities suitable for control and providing insight into the required perturbations [7]. Information obtained from this analysis can significantly narrow the solution space by **selecting features** relevant to the control. Complementary to this **action-space reduction**, the framework employs advanced machine learning techniques to determine low-dimensional dynamical models (**ROM**) that capture the essential flow physics while dramatically reducing the dimensionality of the **environment state space**. This dual reduction approach—**constraining both the action space through stability insights and the state space through dimensionality reduction**—creates a **"dynamically-constrained" domain** where the

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effective application of **symbolic-regression** techniques becomes feasible, enabling the development of **interpretable control strategies** [14].

The **interpretability** of these control laws offers two principal benefits. First, interpretable models **facilitate data assimilation** techniques for real-time adaptation when control efficiency deviates from optimal due to changing external flow parameters [16]. Second, interpretability **supports generalisation** beyond the flow conditions used for development. Through interpretable models, control laws developed from limited numerical datasets can be extended to previously unexplored flow regimes. **The framework will generate interpretable control laws capable of manipulating secondary instabilities with minimal energy input**, significantly enhancing comprehension of the mechanisms driving flow instabilities and their responses to control interventions.

This research will focus on **Görtler vortices—counter-rotating longitudinal vortex pairs** that manifest in boundary layers over concave surfaces due to the imbalance between centrifugal forces and pressure gradient [18]. The methodology encompasses systematic variation of flow regimes across different curvatures, transitioning from configurations where Görtler instabilities arise spontaneously to those where such structures remain absent. In configurations exhibiting natural Görtler vortices, the investigation aims to determine optimal perturbation parameters for effective control, whereas in scenarios where these vortices are not naturally present, the research objective is to identify the minimal perturbation threshold required to induce and manipulate these vortical motions. This approach is designed to establish robust control laws capable of **enhancing momentum mixing efficiency while preserving minimal drag characteristics**.

The selection of this particular flow phenomenon was motivated by its considerable versatility across different flow configurations, which can be achieved without necessitating modifications to the experimental device and set-up. Nevertheless, it should be emphasised that the proposed methodological framework possesses sufficient generality to be applicable to diverse flow scenarios characterised by instability mechanisms. When considering specifically secondary flow phenomena, the methodology may be extended to flows over flat plates with organised roughness elements or riblets [20], non-circular duct flows generating Prandtl's secondary flows of the second kind [4], and junction flows where horseshoe vortical structures develop [3]. Thus, the numerical and experimental approaches developed in this project will have significant implications across a broad spectrum of fundamental flow control applications. The methodology promises significant advancements in both fundamental understanding and practical applications by enabling efficient, interpretable and physically consistent control strategies that exploit intrinsic flow instabilities, thereby addressing critical limitations in current flow control paradigms.

The added value of this project lies in its interdisciplinary nature, bridging the traditionally separate fields of stability theory and machine learning to advance the fundamental understanding of turbulent flow control. The scientific contribution extends beyond specific control strategies to encompass a novel methodological approach for **integrating physics-based insights with data-driven discovery**. The knowledge produced will include not only practical control strategies for specific flow configurations but also generalised principles for **designing interpretable and adaptable control laws** across various fluid dynamics applications. Furthermore, the development of energy-efficient control strategies addresses the need for resource-optimised solutions in engineering applications, representing a significant step towards the rational design of effective control systems with broad applicability across aerospace, energy, and transportation sectors.

b Position of the project as it relates to the state of the art

The proposed **stability-integrated machine learning framework** represents a fundamental advancement in active flow control strategies by synergistically combining physics-based stability models with data-driven techniques. This innovative approach addresses critical limitations in existing methodologies, enabling the development of efficient and interpretable control strategies for complex fluid flows.

Current active flow control approaches using machine learning fall mainly into three primary categories, each presenting **significant limitations**. **Deep Reinforcement Learning (DRL)** methods have demonstrated effectiveness in discovering control policies [6], but demand enormous computational resources and lead to non-interpretable **black-box** models that provide minimal physical insight and limited transferability across flow conditions. **Symbolic Regression** techniques produce interpretable mathematical expressions [14], but become computationally intractable when applied to the high-dimensional spaces characteristic of fluid dynamics problems. Finally, **parameter optimisation** within predefined control laws offers compu-

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tational efficiency but requires prior knowledge of appropriate control functions—knowledge that is rarely available for complex flow scenarios.

The **stability-integrated machine learning framework** introduced in this research project, illustrated in Figure 1, is considered a paradigm shift in machine-learning flow control whereby these limitations are overcome through the effective integration of physical models with data-driven discovery.

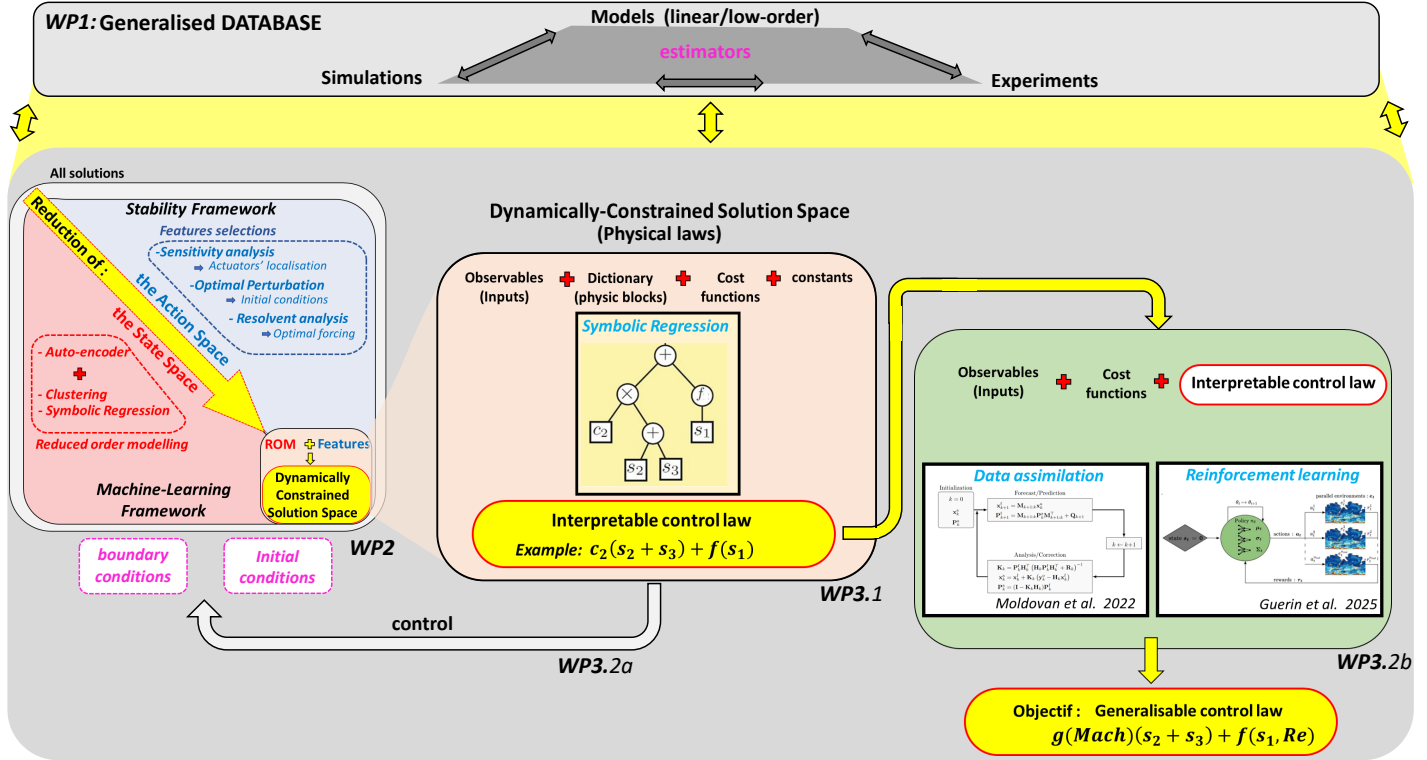


Figure 1. Physical-based stability analysis is combined with data-driven techniques in the stability-integrated machine learning framework to enable interpretable active control laws that leverage flow instabilities.

Model-driven stability analysis is **integrated** with data-driven techniques through four interconnected modules, whereby physical insights are systematically transformed into practical control strategies:

1. **Features selection:** Multiple complementary stability analysis methods—sensitivity analysis, optimal perturbation analysis, and resolvent analysis—are performed to systematically identify the dominant flow instabilities and their sensitivity to perturbations. The control parameter space is **significantly narrowed** by this physics-based approach to dimensions which can be embedded in machine-learning dictionaries. Essential constraints that guide the subsequent machine learning processes are provided by characterising the fundamental instability mechanisms in this module.
2. **Reduced-order modelling:** Low-dimensional manifolds that efficiently capture the essential dynamics of flow structures are extracted by autoencoder architectures [1]. The high-dimensional flow state is transformed by this module into a compact representation where machine learning algorithms can operate effectively.
3. **Interpretable control law discovery:** The constrained parameter space is explored by physics-informed symbolic regression techniques to derive explicit mathematical expressions for control laws. Clear insights into the control mechanisms are provided by these expressions, unlike black-box approaches, and can be efficiently implemented in real-time applications. Mathematical simplicity is balanced with physical fidelity in the resulting control laws, whereby both interpretability and effectiveness are ensured.
4. **Adaptive optimisation:** Integration with data assimilation techniques [16] and reinforcement learning methods [11] is enabled by the interpretable nature of the derived control laws. Optimal control is ensured by this final module even when flow conditions deviate from those used during the initial design phase. Targeted parameter adjustments that preserve the physical principles underlying the

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control strategy while adapting to changing flow conditions are allowed by the explicit mathematical structure.

The present approach has the potential to broaden the current state of knowledge in a number of important aspects, a novel connection is formed through the development of **physics-informed dictionaries for symbolic regression**, wherein physical principles derived from stability analysis are directly incorporated into the mathematical components available for constructing control laws. Through this integration, the symbolic regression search is guided toward physically relevant functions and variables, substantially accelerating convergence toward robust control laws. Additionally, **reduced-order dynamical models** are constructed directly in the low-dimensional latent space identified by autoencoders. This advancement enables efficient real-time prediction and control of complex flow structures while preserving the nonlinear dynamics critical for accurate prediction—a capability not fully realised in current reduced-order modelling techniques. Our framework places specific focus on the **systematic exploitation of secondary instabilities**, particularly centrifugal instabilities in the form of Görtler vortices [7, 18]. Although these structures have been identified in previous research, our framework provides the first comprehensive approach for their targeted manipulation with minimal energy input through active control techniques. The framework further distinguishes itself by producing **interpretable and adaptive control laws** in the form of explicit mathematical expressions. These expressions not only provide physical insight but also enable data assimilation techniques for real-time adaptation to changing conditions [16]—a significant advance over current approaches that typically fail when flow parameters deviate from design conditions. A **transformative approach** to flow control is thus offered by our project through the integration of physical insight with data-driven discovery, whereby more efficient, robust, and physically interpretable solutions than currently possible with either approach in isolation are promised.

c Methodology and risk management

The proposed research program adopts a comprehensive multidisciplinary approach that synergistically combines **model-driven stability analysis**, **data-driven machine learning techniques**, both **low** and **high-fidelity numerical simulations**, and **targeted experimental validation** and generalisation. This integrated methodology enables a robust exploration of flow control strategies that leverage intrinsic instability mechanisms while ensuring interpretability and practical applicability. The program is structured to systematically connect fundamental fluid-dynamics theory with advanced computational methods, creating an innovative framework that transcends the limitations of individual approaches.

The research strategy is organised into interconnected work packages that establish a progressive path from generating foundational knowledge to developing practical control applications, as conveyed by the Gantt shown in table 1. By combining expertise from multiple disciplines—fluid mechanics, stability theory, flow control, and machine learning—the project creates a unique opportunity to address the complex challenges of flow control from complementary perspectives. This cross-disciplinary collaboration is further strengthened through the strategic co-supervision of doctoral researchers, ensuring knowledge transfer across institutional boundaries.

A major innovation in our methodology is the **bidirectional information flow** between physical models and data-driven techniques. Stability analysis provides physical insights that constrain the solution space for machine learning algorithms, while data-driven methods enable the discovery of patterns and relationships that might not be immediately apparent through physics-based approaches alone. This mutually reinforcing interaction is expected to accelerate scientific discovery and lead to more robust control strategies than would be possible with either approach in isolation.

The research program is supported by a team of three doctoral researchers, each focusing on complementary aspects of the framework as detailed in Table 2.

This carefully designed structure ensures comprehensive coverage of the project’s main research areas while fostering collaboration and knowledge exchange between the different research teams. The strategic co-supervision arrangements facilitate seamless integration of expertise across fluid dynamics, stability analysis, experimental techniques, and advanced modelling approaches, thereby enhancing the robustness and impact of the research outcomes.

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Work-Packages	Risk	Investigators			Project Timeline (Quarters)															
		PhD1	PhD2	PhD3	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
WP0: Project Management																				
0.1: Project Management	Low																			
WP1: Multisource database development																				
1.1: Numerical & Experimental Data	Low	X	X																	
1.2: Estimators Development	Med		X																	
1.3: Database Integration	Low	X																		
WP2: Dynamically-constrained solution space																				
2.1: Action Space Reduction	Low/Med	X																		
2.2: State Space Reduction	Low/Med			X																
2.3: Stability Analysis with ML	Med/High			X																
WP3: Generalisable control law development																				
3.1: Interpretable Control Laws	Med		X																	
3.2a: ROMs with Control Effects	High !	X		X																
3.2b: Data Assimilation and Reinforcement Learn.	Med		X																	

Table 1. Project timeline with work packages, tasks, risk levels (Low, Medium, High), and researcher assignments over 48 months.

PhD	Supervisors	Research Focus	WPs and Tasks
1	J.-C. Robinet (DynFluid) L. Larchevêque (IUSTI)	Stability analysis and numerical simulations	WP1, WP2, WP3 (Tasks 1.1, 1.3, 2.1, 3.2a)
2	P. Dupont (IUSTI) P. Traoré (Pprime)	Experiments, numerical simulations with control effects, data-assimilation	WP1, WP3 (Tasks 1.1, 1.2, 3.1, 3.2b)
3	L. Agostini (Pprime) J.-C. Loiseau (DynFluid)	Reduced-order modelling, symbolic regression (SINDy and GEP)	WP2, WP3 (Tasks 2.2, 2.3, 3.2a)

Table 2. Doctoral researchers' assignments and contributions to the research program.

c1 Work package 0 : Project Management

Leader: Lionel Larchevêque (IUSTI)

The macro-management of the project's progress will be carried out through classic six-month progress meetings involving all the partners. The production of associated progress reports (mid-term and final) and other various deliverables will be anticipated by setting up a continuous reporting system based on the PLMLatex collaborative writing tools proposed by the CNRS.

A important condition for the success of the project is that each partner is closely involved in carrying out each work package so that the consortium can benefit from its own expertise. This synergy of knowledge will naturally be based on the network of three doctoral researchers involved in the project, whose activities are all shared between at least two partners. Interactions between doctoral researchers and their distant partners will be based on multiple remote scientific meetings, supplemented by on-site visits, including residencies lasting one to two months.

In the event of difficulties or delays in a task that could affect the progress of other tasks, a committee made up of the coordinator and WP coordinators will meet to define mitigation measures to ensure the success of the project.

In addition to overall coordination of progress between the various workpackages, the WP0 leader will

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be responsible for a number of administrative and management tasks. He will be responsible for drafting the initial and final data management plans, as well as managing the project's knowledge dissemination website and the GitHub repositories for disseminating certain tools developed as part of the project.

c2 Work package 1: Multisource database development for flow control

Leader: **Lionel Larchevêque (IUSTI)**

The framework's foundation is constructed upon a comprehensive **multi-source database** integrating information from complementary approaches to fluid dynamics analysis. This database synthesises data from **experiments**, **high-fidelity** (LES) and **low-fidelity** (RANS) numerical simulations, **model-driven stability** analyses, and **data-driven reduced-order models**. By leveraging the strengths of each methodology, the database provides a rich characterisation of flow dynamics across a range of operating conditions.

Task 1.1: Generation of Numerical and Experimental Flow Data

Objectives: To produce high-quality flow data across various configurations of curved walls with different curvatures, systematically transitioning from scenarios where Görtler instabilities occur spontaneously to those where they are absent.

Work Programme: This task is broken down into two main activities corresponding to the creation of experimental measurement and high-resolution simulation.

The main objective of the database is to provide a description of turbulent flow on a curved wall exploring the widest possible range of dynamics. To achieve this, the **IUSTI** S8 wind tunnel capacity for continuously varying Mach numbers (0.4-0.8) and unit Reynolds numbers (by a factor of 4) will be used. The number of radii of curvature evaluated, due to the cost of the experimental models, will be limited to three, which will have to be chosen meticulously so that the Mach and Reynolds variations allow, for each configuration, to sweep a solution space located on either side of the critical Görtler number.

An activity initiated at the start of the project will therefore aim to define up to three distinct geometries. It will be based on a collaboration between **IUSTI** and **DynFluid** involving a Master 2 student, in which two-dimensional RANS calculations will be used to feed linear stability calculations in turbulent conditions. The latter, whether local or global (using the resolvent approach), will make it possible to estimate, with a reasonable degree of confidence, whether Görtler-type vortices are capable of developing for a given set of Mach-Reynolds-radii of curvature values.

Once the geometrical setups will have been selected and built, velocity fields will be measured using stereo-PIV in crossflow planes, with simultaneous recording of unsteady wall pressure. Stochastic analysis of these synchronized data will allow to derive space-time properties of the flow [17]. Additional measurements will be conducted in horizontal planes downstream of the curved wall, enabling the derivation of transverse wavenumber analyses.

Two to four wall-resolved LES simulations reproducing the experimental conditions for distinct Mach numbers will be carried out at the same time using the open-source FastS code, developed by ONERA, relying on national computational resources provided by GENCI. The corresponding Reynolds numbers will be chosen so that the simulations cover regimes ranging from slightly subcritical to highly supercritical in terms of Görtler-type vortices. Unsteady data on various crossflow and wall-parallel planes complementing the experimental database will be stored on disk. It will enable data assimilation, combined with experimental data, as well as SPOD postprocessing, to be matched to the output of the stability analyses described in **WP2.1**. Both will be useful for task **WP1.3**.

Later in the project, a controlled configuration based on the outputs of **WP3.2b** will be investigated both experimentally and numerically (LES). As a prerequisite to these realisations, it will be necessary to include in the experimental and numerical set-ups control devices adapted to the form of the generalisable control law resulting from this work package. Disturbed configurations will experimentally be generated by integrating Dielectric Barrier Discharge (DBD) plasma actuators upstream of the curved wall. On the numerical side, a simple dielectric model – Suzen-Huang model [19] – will be implemented in the FastS LES solver, taking advantage of **Pprime**'s extensive experience in this area.

Statistical, time-dependent characterisations of the controlled flowfield will be obtained from the measurements through sampling synchronised with the control law at various time delays. From these

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phase-locked PIV measurements, phase-averaged results will enable the reconstruction of a complete space-time sequence of the system. A similar post processing will be applied to unsteady data sampled over various planes from the LES computation.

Deliverables: Comprehensive database of velocity fields, wall pressure distributions, and derived quantities (vorticity, Q-criterion for vortex identification) for multiple flow configurations.

Partners' Contributions: IUSTI will lead this task, performing both numerical simulations and experimental studies. DynFluid will contribute to the definition of simulation parameters to ensure compatibility with stability analyses in WP2, whilst Pprime will advise on the numerical modelling of plasma actuators. The partners will discuss to define the most relevant measurable quantities for constructing the estimators.

Risks and Fall-back Solutions: Potential computational resource limitations may constrain the number of LES cases. Fall-back strategy includes prioritising key cases and supplementing with additional RANS simulations. The different measurement techniques (LDA, PIV, unsteady wall pressure, etc...) are available and well-mastered for high-speed turbulent flows by the IUSTI partner [8, 9, 12]. The plasma actuator systems are currently under development for Mach and Reynolds number conditions similar to this study and will be available at the start of the program. The main experimental challenge will be to accurately design the curved wall from the preliminary RANS-stability analysis.

Task 1.2: Development of Estimators for Real-time Flow Prediction

Objectives: To develop robust estimators that predict the evolution of flow variables of interest (friction, mixing-momentum) from limited measurements accessible in practical applications.

Work Programme: Estimators will be developed using machine learning techniques including neural networks and Gaussian Process Regression. These will be trained to establish mappings between sensor data (wall pressure, point velocity measurements) and global flow states. Both deterministic and probabilistic approaches will be implemented to quantify uncertainty in predictions. The estimators will be benchmarked against full-field data from simulations and experiments from Task 1.1.

Deliverables: A suite of validated estimators for real-time flow prediction with quantified uncertainty, implemented as software modules compatible with control applications.

Partners' Contributions: IUSTI will lead this task, focusing on developing the estimator architectures and training procedures. This work will be carried out by Ph.D. student 2, taking advantage of the tools and expertise of Pprime in data-driven modelling. This collaboration will be achieved through numerous online and onsite exchanges, including a one- to two-month stay at Pprime. DynFluid will advise on essential flow features that must be accurately captured.

Risks and Fall-back Solutions: Potential challenges include insufficient accuracy of estimators for complex flow features. Risk mitigation involves implementing hierarchical estimator structures that progressively increase in complexity, allowing fall-back to simpler but more robust models if needed. Additionally, hybrid physics-informed/data-driven approaches will be developed to improve performance where pure data-driven methods struggle.

Task 1.3: Database Integration and Management

Objectives: To integrate data from multiple sources (simulations, experiments, stability analyses and reduced-order models from WP2) into a structured, accessible database system.

Work Programme: A unified data structure will be developed to standardise the representation of flow data across different sources. These sources will include the raw experimental and numerical database gathered in WP1.1 as well as modal description of the flow obtained from WP2.1 and data built from the reduced-order model developed in task WP2.2. Derived data obtained from the WP1.1 database, conceptually closer to the additions from WP2.1 and WP2.2 than the raw data, will also be added. It will include modal decomposition of the LES flowfields achieved through SPOD and data assimilation from experimental measurements based on estimators built in task WP1.2.

Metadata standards will be established to facilitate search and retrieval based on flow parameters. Quality assessment metrics will be implemented to evaluate data reliability. Additionally, a versioning system

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will be employed to track database evolution as new data is added throughout the project.

Deliverables: Integrated database platform with search capabilities and documented APIs for data retrieval, plus a comprehensive report on data quality assessment.

Partners’ Contributions: IUSTI will lead the database architecture development while coordinating contributions from all partners. DynFluid will provide stability analysis results, and Pprime will contribute to reduced-order model data, both from WP2 activities.

Risks and Fall-back Solutions: Data volume may become unwieldy; selective data compression techniques will be employed if necessary, preserving essential flow features while reducing storage requirements.

The database developed in WP1 serves as a critical foundation for subsequent work packages. By integrating multiple data sources, it enables comprehensive characterisation of flow patterns and instabilities that will directly inform the development of control strategies in later phases of the project. The estimators developed in Task 1.2 will ultimately be integrated with the interpretable control laws derived through symbolic regression, enabling a powerful adaptive control framework, capable of maintaining optimal performance despite changes in flow conditions or uncertainties in the underlying models.

c3 Work package 2: Defining a dynamically-constrained solution space

Leader: Jean-Christophe Robinet (DynFluid)

Task 2.1 : Reducing the Action space – Perturbation Identification and Flow Instability Characterisation

A turbulent wall flow has coherent structures in the form of counter-rotating vortices, their dynamics are based on the fundamental and universal mechanisms of wall turbulence such as the Waleffe cycle for example. On the other hand, the generation of these vortices is specific to the case studied, they can be generated either by geometry, obstacles or curvature for example, or by mode interference mechanisms linked to the non-normality of the evolution operator. The nature of instabilities depends on how they are generated.

Objectives: Control such vortices developing on a concave wall, for example, with the aim of maximizing mixing. To do this, it is necessary to know the instabilities developing within the flow: their nature (convective or not), their dependencies on the control parameters (Reynolds, Mach, Prandtl numbers, etc.) and their spatio-temporal evolutions (spatial localisation, symmetry, wavenumber and frequency). Knowledge of these instabilities will serve as ingredients, as input data, for the development of reduced-order models (Tasks 2.2 and 2.3).

Work Programme: Several tools can be used for the study of instabilities. The DynFluid laboratory has been developing and using several codes (Nek5000 for incompressible flows [10] and dNami for compressible flows [2]) for more than ten years, allowing the computation of the modal and non-modal linear stability of 3D flows. The flow considered in this project is a concave plate where an initially turbulent flow develops centrifugal-type instabilities. These are convective in nature and therefore fall under a receptivity process. **Local stability, global resolvent** and **sensitivity methods** will be used to characterise the spatio-temporal properties of these instabilities.

- The **sensitivity** methods will make it possible to know the **important areas** of the flow for the development of these instabilities,
- the **resolvent** will make it possible to know the **optimal forcing**,
- the **frequency selectivity** of the flow and the **linear stability** will determine which are the **unstable modes** of the flow.

Deliverables: This task aims to provide a better understanding of the physical mechanisms responsible for the appearance of centrifugal instabilities and data to feed and guide the learning methods of Tasks 2.2 and 2.3.

Partners’ Contributions: DynFluid will lead the study of receptivity, stability and sensitivity in interaction with IUSTI in particular the nonlinear simulations allowing to validate the relevance of the stability study.

Risks and Fall-back Solutions: Calculating the global linear stability around a turbulent mean field is

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a delicate task. In order to reduce the risks, several less risky methods (local and resolvent) will be used. In parallel, decomposition methods (SPOD, DMD) from nonlinear simulations will be used and compared to stability methods, they could be fallback solutions if stability methods are ineffective.

Task 2.2 : Reducing the state space – building Reduced-Order Models

Objectives: To develop reduced-order models (ROMs) that efficiently capture the essential dynamics of the flow while significantly reducing computational complexity, enabling real-time implementation of control strategies.

Work Programme: **Autoencoders** are neural network architectures consisting of an encoder that maps high-dimensional input data to a low-dimensional latent space and a decoder that reconstructs the original data from this compressed representation. By minimising reconstruction error during training, autoencoders learn to identify the most important features of the dynamic.

Three complementary methodologies will be implemented for the modelling in the latent space, from the easiest to the more complicated, limiting the risk :

- **Deterministic neural network-based approach:** Convolutional neural networks (CNNs) (see [1]) or transformer architectures will be employed to learn the dynamics directly in the latent space. These models offer **excellent predictive capability but operate as "black boxes"**, limiting interpretability of the underlying physics.
- **Probabilistic clustering approach:** Flow states in the latent space will be partitioned into distinct clusters representing similar dynamical regimes. This technique produces a **coarse-grained representation** of the dynamics, characterising the probability of transitions between states. This approach enables identification of preferable flow states and development of control strategies to promote these states. Though **less precise than neural network models**, this method provides **greater interpretability** of the system's dynamical behaviour.
- **Symbolic regression approach:** **Mathematical expressions** describing the dynamics will be discovered through two complementary methods. First, evolutionary algorithms such as genetic programming (GEP) will be used, which combine basic functions (addition, multiplication, etc.) with relevant flow variables through operations inspired by natural selection. Second, the **SINDy** (Sparse Identification of Nonlinear Dynamics) method will be employed, which leverages sparse regression techniques to identify the most important terms in a predefined library of candidate functions. For fluid mechanics systems with inherently high dimensionality, these symbolic regression approaches would be computationally intractable without careful constraints. Therefore, the search process will be **critically guided by physical insights from stability analysis**, which substantially reduces the dimensionality of the problem and defines a physically-relevant subspace where symbolic regression becomes feasible. This **physics-based constraining of the solution space is essential** for both computational efficiency and ensuring convergence towards expressions that capture the fundamental mechanisms governing the flow dynamics.

Deliverables: A suite of validated reduced-order models with quantified accuracy metrics, comparison reports against full-order simulations, and software implementation for integration with control algorithms.

Partners' Contributions: **Pprime** will lead this task, focusing on autoencoder development and clustering techniques [1, 13]. **DynFluid** will contribute expertise on embedding physically relevant flow features, identified through stability analysis, and participate in the implementation of symbolic regression. Two complementary approaches to symbolic regression will be implemented to mitigate methodological risks: SINDy algorithms [5, 15] (**DynFluid**) and genetic programming (GEP) [14] (**Pprime**). SINDy offers rapid convergence through sparse regression once an appropriate dictionary of basis functions is established, but requires that all potential terms in the target expression be included in this dictionary. Genetic programming provides greater flexibility by constructing expressions through evolutionary processes that can develop complex combinations from simpler building blocks, though at higher computational cost. This dual approach ensures robustness against uncertainty in the underlying physical mechanisms; if SINDy struggles due to missing physics in its predefined dictionary, genetic programming can potentially discover alternative mathematical formulations not initially considered. For instance, if the optimal control law has the form $f(x) = ax + bx^2$, SINDy would require both x and x^2 terms in its dictionary but would converge rapidly

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by identifying the coefficients a and b . In contrast, genetic programming could potentially discover this quadratic relationship even if only x were included in its initial dictionary, by evolving the expression to incorporate the x^2 term through its combinatorial operations.

Risks and Fall-back Solutions: The primary risk is found in the balance between accuracy and interpretability across different ROM approaches. To mitigate risks associated with symbolic regression, both SINDy and genetic programming approaches are employed in parallel—if convergence cannot be achieved with one method due to dictionary limitations or algorithmic constraints, success might be found through different mathematical pathways with the alternative approach. If symbolic regression is proven excessively challenging despite this dual methodology, emphasis will be shifted to the probabilistic clustering approach, which is offered as a middle ground between interpretability and computational tractability. A coarse-grained but physically interpretable representation of the flow dynamics through state transitions is provided by the clustering method, enabling effective control strategies even when precise mathematical expressions cannot be identified. Additionally, the clustered state-space model can be enhanced with physical consistency through insights gathered from partially successful symbolic regression attempts.

Task 2.3 : Coupling Stability Analysis with Machine Learning

Objectives: To develop a novel integration framework that combines physical insights from stability analysis with machine learning approaches, thus creating physics-informed dictionaries for symbolic regression and constraining the search space for the discovery of control laws.

Work Programme: This task will establish the methodological connection between physics-based and data-driven approaches. Results from stability analyses ([Task 2.1](#)) will be systematically transformed into constraints and features for machine learning algorithms. This process involves: (i) Identifying the most sensitive flow regions and dominant instability mechanisms from stability analysis. (ii) Translating these physical insights into mathematical components for symbolic regression dictionaries. (iii) Developing filtering criteria based on physical principles to reject implausible control laws during the evolutionary process of symbolic regression. (iv) Creating a formal mapping from the high-dimensional flow description to the reduced-order state space.

The resulting "dynamically-constrained" solution space will significantly reduce the dimensionality of the control problem while preserving the essential physics, thereby enabling efficient application of machine learning techniques.

Deliverables: Physics-informed dictionaries for symbolic regression, formal specifications for the dynamically-constrained solution space, and documentation of the methodological framework for integrating stability analysis with machine learning.

Partners' Contributions: [DynFluid](#) and [Pprime](#) will jointly lead this task, with [DynFluid](#) providing expertise on stability analysis interpretation and [Pprime](#) contributing knowledge on machine learning implementation. [IUSTI](#) will advise on practical considerations for experimental validation.

Risks and Fall-back Solutions: The primary risk lies in the potential mismatch between linearised stability analysis predictions and actual nonlinear flow behaviour. To mitigate this risk, iterative refinement of the constraints will be implemented, with initial control laws tested against full nonlinear simulations to identify discrepancies. If the integrated approach proves challenging, a more loosely coupled strategy will be adopted, where stability analysis and machine learning operate semi-independently with structured information exchange rather than full integration. This work package represents the core methodological contribution of the project, establishing the novel framework for combining physics-based insights with data-driven discovery. The dynamically-constrained solution space created here will directly enable the development of interpretable control laws in subsequent work packages, while ensuring that these laws remain physically consistent and computationally tractable.

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c4 Work package 3: Generalisable control law leveraging secondary flow instabilities

Leader: **Lionel Agostini (Pprime)**

This work package represents the culmination of the research programme, where insights from stability analysis and reduced-order modelling are synthesised into practical control strategies. The package is divided into two complementary tasks, each addressing a distinct aspect of the control problem.

Task 3.1: Design of Interpretable Control Laws

Objectives: To develop interpretable control laws that effectively manipulate Görtler vortices for specific flow configurations, enhancing mixing momentum.

Work Programme: Building upon the dynamically-constrained solution space defined in WP2, symbolic regression techniques will be employed to derive explicit mathematical expressions for control laws (SINDy and GEP). The focus will be placed on control laws that exploit the inherent instability mechanisms of the flow, requiring minimal energy input for significant flow modification. The control laws will be evaluated through numerical simulations and experimental testing, with iterative refinement based on performance metrics.

For naturally occurring Görtler vortices, control strategies will be developed to modify their strength, spacing, and wall-normal extent. In configurations where Görtler vortices are not naturally present, minimal-energy actuation strategies will be designed to trigger and subsequently manipulate these structures. Across all scenarios, the objective function will be to enhance the mixing momentum.

Deliverables: A library of interpretable control laws for different but limited flow configurations, accompanied by detailed performance analyses.

Partners' Contributions: Pprime will lead this task, with contributions from DynFluid on stability-informed control strategies and from IUSTI on experimental implementation considerations.

Risks and Fall-back Solutions: The primary risk involves the potential complexity of optimal control laws, which may resist expression in simple mathematical forms. If fully interpretable expressions cannot be identified for certain flow regimes, a hierarchical approach will be adopted, whereby simplified expressions are employed for the dominant mechanisms, supplemented by black-box components for secondary effects.

Task 3.2: Development of Adaptive Control Frameworks

Objectives: To extend the applicability of the control laws beyond their design conditions through two complementary approaches: (1) **development of reduced-order models with embedded control effects**, and (2) **implementation of data assimilation and reinforcement learning techniques for real-time adaptation**.

Work Programme: Two parallel strategies will be pursued:

- **Task 3.2.a**– The first approach involves the development of reduced-order models that incorporate control effects directly. These models will be constructed by selecting flow features identified through stability analysis that are most relevant to specific control objectives. Machine learning algorithms will be employed to assist in this feature selection process. The resulting ROMs will enable prediction of flow behaviour under varying boundary conditions, initial conditions, and control inputs. This approach, whilst ambitious, offers the potential for purely model-driven adaptation of control strategies to new objectives without requiring additional simulation or experimental data.
- **Task 3.2.b**– The second, more conservative approach, leverages the interpretability of the control laws developed in Task 3.1. Data assimilation techniques will be implemented to update model parameters in real-time based on sensor measurements [16], while reinforcement learning methods will be employed to optimise control parameters for changing flow conditions. For the reinforcement learning, policy-based optimisation will be used [11]. This approach is already implemented within Pprime simulation code and will be ported to the experiments. This strategy adapts the control law directly without necessitating complete flow modelling, providing a robust alternative to the more ambitious ROM approach.

Deliverables: Reduced-order models with embedded control effects for selected flow scenarios, plus data assimilation and reinforcement learning frameworks for adaptive control implementation.

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Partners’ Contributions: Pprime and Dynfluid will work hand-in-hand to the development of ROMs with embedded control effects. Pprime and IUSTI will closely collaborate on the reinforcement learning aspects and data assimilation framework.

Risks and Fall-back Solutions: The development of ROMs with embedded control effects represents a significant scientific challenge. If this approach proves intractable for the complex flow physics under consideration, emphasis will be shifted entirely to the data assimilation and reinforcement learning strategy, which offers a more conservative but reliable path to adaptable control. The modular structure of the research programme ensures that valuable outcomes will be achieved even if the most ambitious objectives cannot be fully realised.

This work package builds directly upon the foundations established in WP1 (multisource database) and WP2 (dynamically-constrained solution space), transforming theoretical insights into practical control strategies. The dual approach to adaptive control mitigates research risk while pursuing significant scientific advances, ensuring both practical outcomes and potential breakthroughs in flow control methodology.

c5 Deliverables

The deliverables associated with the various activities described in each of the four work packages are listed in Table 3, along with the corresponding schedule.

Delivery date	WP	Item	Title	Nature
$t_0 + 3$	WP0	D0.1	Project website and initial data management plan	Website, report
$t_0 + 6$	WP1	D1.1	Geometric definition of the curved models	Technical report
$t_0 + 18$	WP1	D1.2	Uncontrolled database (experiments+LES)	Database
$t_0 + 18$	WP2	D2.1	Linear Stability Analysis (Global resolvent)	Technical report
$t_0 + 20$	WP1	D1.3	First version of estimators for flow prediction	Technical report
$t_0 + 24$	WP2	D2.2	Initial reduced order models	Technical report
$t_0 + 24$	WP0	D0.2	Mid-term report	Report
$t_0 + 28$	WP2	D2.3	Physics-informed dictionary for symbolic regression	Technical report
$t_0 + 28$	WP1	D1.4	Consolidated uncontrolled DB (data assim. & stability)	Database, report
$t_0 + 42$	WP1	D1.5	Optimized controlled database (experiments+LES)	Database
$t_0 + 42$	WP3	D3.1	ROM with embedded control effects	Technical report
$t_0 + 46$	WP3	D3.2	Data assimilation and reinfor. learning framework	Technical report
$t_0 + 48$	WP0	D0.3	Final report and final data management plan	Report

Table 3. Deliverables of the project.

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II Organisation and implementation of the project

a Scientific coordinator and its consortium / its team

The project BENEFIT is coordinated by **Lionel Larchevêque (IUSTI)**. It is organized in three technical WPs divided in tasks. Each WP is under the responsibility of a well-identified coordinator namely **L. Larchevêque (IUSTI)** for WP1, **J.-C. Robinet (DynFluid)** for WP2, and **L. Agostini (Pprime)** for WP3.

Coordinator and workpackage leaders:

L. LARCHEVÊQUE, (Maître de conférence), is member of the supersonic group of the **IUSTI** laboratory (UMR 7343 CNRS Aix-Marseille University). He will contribute to Tasks 0, 1.1, 1.3 and 3.2b and will be involved in the co-supervision of PhD student 1 with **DynFluid**. His research focus on high-fidelity simulations of compressible flows complementing experimental database in view of advanced joint-analyses of the dynamics of separated flows. He was involved in 4 French/European research projects and has co-supervised 5 PhD students.

J.-C. ROBINET Dr.-Ing., full professor at Arts et Métiers Institute of Technology, head of Fluid Dynamics Laboratory (**DynFluid**), academic staff, state funding. He will be mainly involved in Tasks 2.1, 1.1 and 1.3. He will also co-supervise PhD student 1 with **IUSTI**. Jean-Christophe Robinet is an expert in compressible flows in the field of aeronautics and aerospace. He is a recognized specialist in the development of methods and the study of instabilities and the transition to turbulence. He has participated in 7 ANR projects since 2007 and one European project.

L. AGOSTINI (HDR, Chargé de Recherche CNRS at **Pprime**) has significant expertise in wall-bounded turbulent flows, drag reduction strategies, and machine learning applications for fluid dynamics. His research on autoencoder technologies for dimensional reduction and reinforcement learning approaches for flow control has demonstrated promising results in recent studies. His work on data-driven modelling approaches will be valuable for the project, particularly in developing interpretable reduced-order models that balance computational efficiency with physical consistency. He will lead WP3 and contribute to tasks involving reduced-order modelling, reinforcement learning implementation, and control law development, helping to integrate stability theory insights with machine learning methodologies.

The implications of the coordinator and the leaders of the work packages in currently active research projects are listed in Table 4.

Researcher	Person.month	Call, funding agency, grant allocated	Project's title	Scientific coordinator	Start–End
J.-C. Robinet	8	ANR-PRC	IDEFHYX	N. Mazellier	2023-2027
	8	DGAC	MAMBO	G. Pont	2021-2025
	5	ANR-JCJC	NINJA	I. B. H. Saïdi	2024-2028
L. Agostini	39	ANR-JCJC	INFERENCE	L. Agostini	2024-2028
	6	ANR-PRC	SOLAIRE	A. Toutant	2022-2025
	4	ANR-PRC	MUFDD	L. Perret	2023-2027

Table 4. Implication of the scientific coordinator and partner's scientific leaders in on-going project(s).

Other project members:

S. PIPONNIAU (Maître de Conférence at **IUSTI**) is a specialist in compressible flows and experimental methods applied to high-speed flows. He is responsible for the Supersonic Wind Tunnel at **IUSTI**, which has been designated as the "Compressible Flows" research platform of Aix-Marseille University. His research focuses on the identification of turbulent structures through modal decompositions, conditional approaches, or stochastic estimations. He was involved in 3 European research projects and has co-supervised 3 PhD students. He will be involved in Tasks 1.1, 1.3 and 3.2b.

P. DUPONT (HDR, Chargé de recherche CNRS at **IUSTI**). He is a specialist in compressible flows. He was Deputy Director of **IUSTI** from 2018 to 2022, he was the scientific lead for **IUSTI**'s contributions in three

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European programs and coordinated the ANR DECOMOS (2010-2014). He is also responsible for research programs in collaboration with CNES and has been a member of the steering committee (COFIL) for establishing a CNRS network on large-scale experimental facilities (Réseau RAMSES). He will co-supervise PhD student 2 and contribute to tasks 1.1, 1.3, and 3.2b, where his expertise in high-speed turbulent flow experiments will be especially beneficial.

P. TRAORÉ (Professor at **Institut Pprime**) is a specialist in numerical simulation methods for complex flows and has developed the in-house code ORACLE3D for fluid mechanics simulations with plasma actuators. His expertise in control strategies and numerical modelling will support multiple aspects of the research programme. He will co-supervise PhD student 2 and contribute significantly to Tasks 1.1, 3.1, and 3.2b.

L. CORDIER (Directeur de Recherche CNRS at **Institut Pprime**) brings critical expertise in dimensional reduction techniques, data-driven modelling and data assimilation. His experience with reduced-order models and flow control will be particularly valuable for WP2 and WP3. He will contribute to Tasks 2.2, 2.3, and 3.2a.

F. KERHERVÉ (Maître de Conférences at **Institut Pprime**) is specialised in experimental fluid mechanics and signal processing. He will contribute to estimator development, data assimilation, and reinforcement learning implementation, helping numerical and experimental components of the project. He will be involved in Tasks 1.2, 3.2b, and lead the adaptation of reinforcement learning frameworks from simulations to experimental platforms.

J.-C. LOISEAU, Assistant professor (**DynFluid**), state funding since September 2018. He will be mainly involved in the Tasks 2.2, 2.3 and 3.2a. He defended his PhD thesis on hydrodynamic stability analyses of 3D flows in 2014. In 2018, he obtained a position as assistant professor at Arts et Métiers. His research activities are mainly: (1) elucidating the physical mechanisms responsible for transition to turbulence in 3D flows, and (2) the development and use of data-driven techniques for reduced-order modelling in the physical and engineering sciences.

N. ALFEREZ, Assistant professor (**DynFluid**), state funding since September 2018. He will be mainly involved in the Tasks 2.1, 1.1 and 1.3. He defended his PhD thesis on LES of the stall mechanism induced by the bursting of a Laminar Separation Bubble in 2014. From 2014 to 2016 he obtained a research associate position at Imperial College of London on shock refraction properties in dense gases. From 2016 to 2018 he obtained a second research associate position at ONERA in collaboration with Intel Parallel Computing Center on code modernization. In 2018, he obtained a position as assistant professor at CNAM. Within **DynFluid**, he develops the dNami code, an open-source multi-language framework for solving systems of balance laws using explicit numerical schemes on structured meshes, in particular for instability problems in compressible regimes thanks to automatic differentiation methods.

b Implemented and requested resources to reach the objectives

Partner 1: Institut Universitaire des Systèmes Thermiques Industriels -IUSTI

Staff expenses

Academic staff: 1 assistant professor (18 person/month) (**L. LARCHEVÊQUE**), 1 assistant professor (8 person/month) (**S. PIPONNIAU**), 1 Chargé de Recherche HC (8 person/month) (**P. DUPONT**), 1 Ingénieur d'études HC (8 person/month) (**P. LANTOINE**).

1 PhD student (36 months)

Staff Expenses			
1	PhD student (36 person/month)	:	120,000.00 €
Sum		:	120,000.00 €

Instruments and material costs The partner requires 36,500 € of Instruments and material fundings. They correspond to the realisation of the experimental set-up (curved walls, plasma-actuators slot, unsteady wall pressure sensors, DBD fabrications). Most of the metrology systems required for this project are available (PIV, LDA, signal conditioners etc...). The PIV system will require to be upgraded for the stereoscopic measurements (8000 € are asked to contribute to the acquisition of a 9MPx camera and the

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associated scheimpflug). The last expense corresponds to the acquisition of a laptop for the PhD student and of a minimal post-processing workstation for the PIV experiments.

Outsourcing / subcontracting This project will require about 20 hours of the IUSTI's high speed wind tunnel (500 €/h for collaborative research programmes). This corresponds to 10 k€ (internal invoice from the AMU's platform Compressible Flows).

Overheads costs (*Frais généraux non-forfaitisés* in French) The project involves three permanent researchers and one PhD student, all of whom will require travel support. We request €10k€ for the overhead costs, allocated as follows:

- 2 k€ for consortium 6-month meetings participation
- 4 k€ for international conferences to promote and disseminate the results obtained during the project
- 4 k€ for PhD student mobility. This will allow him/her to benefit from stays at the Pprime and Dynfluid partners to further his/her research in data assimilation and stability analysis.

All expenses of partner **IUSTI** not related to staff are summarized in Table 5.

IUSTI Laboratory										
Instruments and material costs					Overheads costs:					
1	Experimental Setup	:	24,500	€	10	Consortium meeting travels	:	2,000	€	
1	Stereo-PIV Components	:	8,000	€	2	Intern. conferences	:	4,000	€	
1	Computers	:	4,000	€	1	PhD Mobility	:	4,000	€	
Sum		:	36,500	€					10,000	€
Outsourcing / subcontracting										
20	Wind tunnel Operation	:	10,000	€						
Sum		:	10,000	€						

Table 5. Summary of resources requested for IUSTI laboratory.

Partner 2: Pprime

Staff expenses

Academic staff: 1 full professor (10 person/month) (**P. Traoré**), 1 assistant professor (8 person/month) (**F. Kerkervé**), 1 Chargé de Recherche (10 person/month) (**L. Agostini**) and 1 Directeur de Recherche (8 person/month) (**L. Cordier**).

1 Master students (6 months)

1 PhD student (36 months)

Staff Expenses			
1	Master student (6 person/month)	: 4,300.00 €	
1	PhD student (36 person/month)	: 127,000.00 €	
Sum		: 131,300.00 €	

Instruments and material costs The partner requires €59,700 for computing infrastructure and associated equipment, allocated as follows:

- €46,000 for a dedicated hybrid CPU/GPU cluster optimized for the integrated computational workflows central to this research,
- €3,000 for laptops to equip PhD students and facilitate mobile research activities,
- €700 for monitors to enhance workstation productivity,
- €10,000 for storage solutions to manage the substantial data volumes generated by numerical simulations.

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This investment complements our existing computing infrastructure, which includes two local hybrid clusters that have proven highly effective for developing reinforcement learning frameworks and genetic programming approaches for control applications. The hybrid architecture—allowing parallel CPU-based simulations to interact efficiently with GPU-accelerated machine learning algorithms—has been particularly valuable for our autoencoder-based reduced-order modelling work.

The requested new cluster will operate in parallel with existing resources, expanding our computational capacity to meet the increased demands of the multiscale modelling and optimisation tasks specified in WP 2 and 3. For large-scale production runs requiring more substantial resources, we will utilise the regional computing facilities and dynamic access to the national Jean Zay GPU cluster. This tiered approach to computing resources ensures efficient development of algorithms locally while enabling scalability for intensive computational tasks.

Overheads costs Travel support for maintaining scientific engagement and fostering collaboration will be required for four permanent researchers and one PhD student involved in the project. A total of €25,500 for overhead costs is requested, allocated as follows:

- €7,000 for staff conference participation, with €3,500 for international conferences and €3,500 for European conferences. These will be focused on both fluid mechanics and the rapidly growing field of machine learning for fluid dynamics. Regular participation in specialised workshops and conferences is considered essential for maintaining awareness of methodological advances and establishing collaborations with leading research groups in this emerging domain.
- €2,500 for consortium meeting travel, through which essential knowledge transfer between research teams at **Pprime**, **DynFluid** and **IUSTI** laboratories will be supported.
- €6,000 for organising an international symposium on physics-informed machine learning for flow control in the third year of the project, where researchers from fluid dynamics, control theory, and machine learning communities will be brought together to foster interdisciplinary collaboration.
- €10,000 for PhD-specific travels, which is broken down into €2,000 for international conferences, €3,000 for European conferences, €1,000 for summer schools, and €4,000 for extended research mobility visits. These activities are considered critical for doctoral training and for facilitating hands-on collaboration on experimental work, code development, and integration of computational workflows.

This travel budget has been carefully optimised to ensure essential scientific exchange while the resources available for core research activities are maximised. The emphasis on PhD mobility reflects the commitment to training the next generation of researchers at the intersection of fluid dynamics and machine learning.

Table 6 lists the instrument, material and overhead costs of Partner **PPrime**, including the ones specific to the mobility of the PhD student.

Pprime					
Instruments and material costs			Overheads costs		
1	Hybrid CPU/GPU cluster	: 46,000 €	6	Consortium meeting travels	: 2,500 €
3	Laptops	: 3,000 €	2	Permanent intern. conferences	: 3,500 €
2	Monitors	: 700 €	3	Permanent European conferences	: 3,500 €
1	Storage solutions	: 10,000 €	1	International symposium organisation	: 6,000 €
			Overheads costs: PhD-specific travel		
			3	PhD intern. conferences	: 2,000 €
			2	PhD European conferences	: 3,000 €
			1	Summer schools	: 1,000 €
			1	PhD mobility	: 4,000 €
Sum			: 59,700 €		
			: 25,500 €		

Table 6. Summary of resources requested for Pprime Institute (Total: 85,200€).

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Partner 3: DynFluid

Staff expenses

Academic staff: 1 full professor (10 person/month) (J.-C. Robinet), 1 assistant professor (8 person/month) (J.-C. Loiseau), 1 assistant professor (8 person/month) (N. Alferez).

1 PhD student (36 months)

1 Master student (6 months)

Staff Expenses			
1	Master student (6 person/month)	:	4,200.00 €
1	PhD student (36 person/month)	:	130,032.00 €
Sum		:	134,232.00 €

Instruments and material costs The DynFluid laboratory obtains 30Mh on average every year from GENCI's national resources (<https://genci.fr/en>). GENCI is an organization bringing together three national computing centers: CINES (France Universités), IDRIS (CNRS) and TGCC (CEA). The high-performance computers operated at IDRIS (Institut du développement et des ressources en informatique scientifique), mainly an HPE SGI 8600 'Jean Zay' (<http://www.idris.fr/eng/jean-zay/>) and at TGCC (Très Grand Centre de Calcul) a Bull Sequana SKL 'Irène Joliot-Curie' (<https://www-hpc cea.fr/en/Joliot-Curie.html>). Additional computing resources can be requested from PRACE if needed. In addition, DynFluid has a Linux network of 9 servers with 52 Intel Xeon computing cores each, 380 GB of RAM memory and 8 TB of storage which are available for the present project as well.

For the project, it will be necessary to purchase a computer server with around fifty additional cores in order to carry out all the intermediate computations (meshing, validation, development computations, ...). This machine will mostly be reserved for the doctoral students requested. DynFluid has a modular storage server. In order to store all the data from the project, 40 Tb would be enough. An additional memory module for the storage server may be purchased. The laptop will be purchased for the doctoral students who will be recruited by the project.

Overheads costs: For Dynfluid, the number of permanent and non-permanent (1 doctoral student to recruit) people who will participate in this project will be three (Robinet, Loiseau, Alferez and doctoral student). The forecast for mission expenses in each laboratory is presented in Table 7. These are Paris-Poitiers and Paris-Marseille return trips for three people for three days once a year. Three conferences outside Europe for the doctoral student (APS, AIAA), three conferences in Europe (ERCOFTAC, EUROMECH) for the doctoral student and a permanent staff member are also budgeted.

The instrument, material and overhead costs for Partner DynFluid are detailed in Table 7

DynFluid Laboratory					
Instruments and material costs			Overheads costs:		
1	Computer station	: 11,000 €	9 × 2	Paris-Poitiers/Marseille trips	: 4,000 €
1	40 Tb storage disk	: 2,000 €	3 × 2	Intern. conferences	: 9,000 €
1	Laptop	: 2,000 €	3 × 1	Europ. conferences	: 4,000 €
Sum		: 15,000 €			: 17,000 €

Table 7. Summary of resources requested for DynFluid.

Summary of requested resources

The financial requests by partner and by expenditure item are summarised in Table 8, which also shows the total funding requested for the project.

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	Partner IUSTI	Partner Pprime	Partner DynFluid
Staff expenses (non permanent)	120,000.00€	131,300.00€	134,232.00€
Instruments and material costs	36,500.00€	59,700.00€	15,000.00€
Building and ground costs	0€	0€	0€
Outsourcing / subcontracting	10,000.00€	0€	0€
Overheads costs ¹	10,000.00€	25,500.00€	17,000.00€
Administrative management & structure costs	23,827.50€	29,227.50€	22,441.32€
Sub-total	200,327.50€	245,727.50€	188,673.32€
Requested	634,728.32€		

Table 8. Requested means by item of expenditure and by partner.

III Impact and benefits of the project

The **stability-integrated machine learning framework** for flow control developed in this project will generate impacts across multiple domains, from fundamental fluid mechanics to industrial applications, creating both scientific advances and societal benefits.

a Scientific and societal impact

The proposed research will significantly advance the state of the art in **three interconnected fields**. In **fluid mechanics**, the project will deepen understanding of secondary flow instabilities, particularly Görtler vortices, and their manipulation through minimal energy input. This knowledge will contribute to the fundamental science of turbulent structures and their control. In the field of **machine learning for fluid dynamics**, the framework establishes a new paradigm for integrating physical models with data-driven discovery, addressing critical limitations in current approaches that rely exclusively on either physics-based or pure machine learning methods. Finally, in **control theory**, the interpretable control laws developed will expand possibilities for robust control in complex nonlinear systems.

The research will produce **methodological innovations** of broad applicability beyond the specific flow configurations studied. By demonstrating how stability analysis can effectively constrain machine learning solution spaces, the project offers a blueprint for other complex physical systems where interpretable control is desired but direct modelling is challenging. The scientific insights generated will be disseminated through publications in high-impact journals spanning fluid mechanics (*Journal of Fluid Mechanics*, *Physics of Fluids*), machine learning (*Journal of Machine Learning Research*), as well as presentations at leading international conferences such as the American Physical Society’s Division of Fluid Dynamics (APS-DFD) meetings, EUROMECH Fluid Mechanics Conferences, and the International Symposium on Turbulence and Shear Flow Phenomena.

The interdisciplinary nature of the project provides an exceptional training environment for **undergraduate students** and the **doctoral researchers** involved. By working at the intersection of fluid mechanics, stability theory, and machine learning, these Phd students will develop unique expertises that are increasingly valuable in both academic and industrial settings.

b Dissemination and exploitation strategy

The project adopts a comprehensive approach to research dissemination and exploitation:

1. **Open science practices:** In addition to journal publications, all data and computational tools developed will be made accessible through open repositories. The research team will implement a data management plan within six months of project initiation, with datasets published in domain-specific repositories with persistent identifiers. Open-source implementations of key algorithms will be published on platforms such as GitHub with appropriate documentation.
2. **Academic community engagement:** The project will host an international symposium on physics-informed machine learning for flow control in the third year, bringing together researchers from fluid

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dynamics, control theory, and machine learning communities to foster interdisciplinary collaboration and accelerate scientific progress.

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